

Output 1: Unsmoothed N-gram Language Model

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      N-GRAM LANGUAGE MODEL
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1. Unigram
2. Bigram
3. Trigram
4. Predict next word
5. Test unseen N-gram
6. Evaluate model
7. Exit
Enter your choice: 4
Enter N (1, 2 or 3): 2
Enter incomplete sentence: The student is

Top-5 predictions:
reading -> Probability: 0.1667
writing -> Probability: 0.1667
learning -> Probability: 0.1667
studying -> Probability: 0.1667
explaining -> Probability: 0.1667
```

Output 2:

```
1.Unsmoothed 2.Backoff 3.Interpolation 4.Compare 5.Exit
Choice: 2
Enter sentence: I am a student from
Prediction: ('the', 0.16393442622950818)
Used: Unigram

1.Unsmoothed 2.Backoff 3.Interpolation 4.Compare 5.Exit
Choice: 3
Enter sentence: Pizza is my favourite
Prediction: ('the', 0.03278688524590164)

1.Unsmoothed 2.Backoff 3.Interpolation 4.Compare 5.Exit
Choice: 4
Enter sentence: Hi how are you
unsmoothed : ('<s>', 0)
backoff : ('the', 0.16393442622950818)
interpolation : ('the', 0.03278688524590164)
```

Output 3:

```
N = 1
<s> the student is reading a book </s> -> 3.9295248433727457
<s> the student likes machine learning </s> -> 3.876457731694272
<s> the teacher is teaching the student </s> -> 3.560244486126866
<s> the student plays football </s> -> inf

N = 2
<s> the student is reading a book </s> -> 0.42857142857142855
<s> the student likes machine learning </s> -> 0.5974937501201927
<s> the teacher is teaching the student </s> -> 1.024275000206044
<s> the student plays football </s> -> inf

N = 3
<s> the student is reading a book </s> -> inf
<s> the student likes machine learning </s> -> inf
<s> the teacher is teaching the student </s> -> inf
<s> the student plays football </s> -> inf

--- Smoothed Entropy ---
N = 1 Average = 3.8862
N = 2 Average = 2.9428
N = 3 Average = 3.3571

Interpretation:
Low entropy = more predictable words.
High entropy = less predictable words.
Smoothing avoids zero probabilities for unseen N-grams.
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Output 4:

1.Rule Based 2.Stochastic 3.Transformation 4.Compare 5.Exit
Choice: 1

Enter sentence: I am sick

Rule Based : i/NN am/NN sick/NN

1.Rule Based 2.Stochastic 3.Transformation 4.Compare 5.Exit
Choice: 2

Enter sentence: This is a beautiful peacock

Stochastic : this/VBZ is/VBZ a/VBZ beautiful/VBZ peacock/VBP

1.Rule Based 2.Stochastic 3.Transformation 4.Compare 5.Exit
Choice: 3

Enter sentence: I like to run daily

Transformation : i/NN like/NN to/NN run/NN daily/RB

1.Rule Based 2.Stochastic 3.Transformation 4.Compare 5.Exit
Choice: 4

Enter sentence: Nice to meet you MR Trump

Rule Based : nice/NN to/NN meet/NN you/NN mr/NN trump/NN

Stochastic : nice/VBZ to/VBZ meet/VBZ you/VBZ mr/VBZ trump/VBP

Transformation : nice/NN to/NN meet/NN you/NN mr/NN trump/NN

1.Rule Based 2.Stochastic 3.Transformation 4.Compare 5.Exit
Choice: 5

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